

Digital signatures

Cryptography

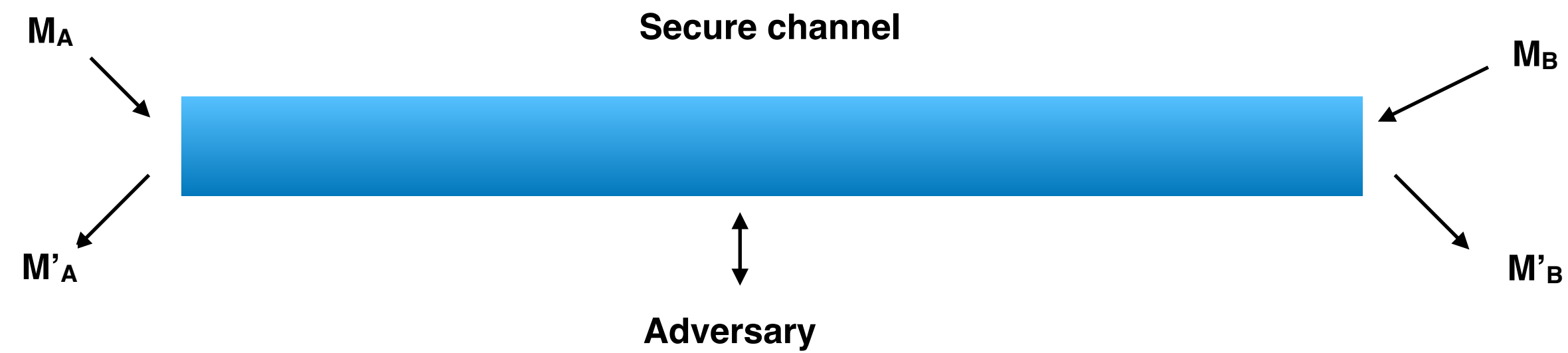
Identity

Who are “Alice” and “Bob”?

Alice



BOB



Names are bound to people by social means

Alice

Alice in wonderland



Alice Waters



Alice Walker



Alice Poker



BOB



Bob squarepants



Bob the builder



Bob Marley



Bob Marley

Cryptography

Identity

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Established out-of-band

Biding of name to a person

Cryptography

Identity

Identity in the internet

These are difficult issues

Established how?

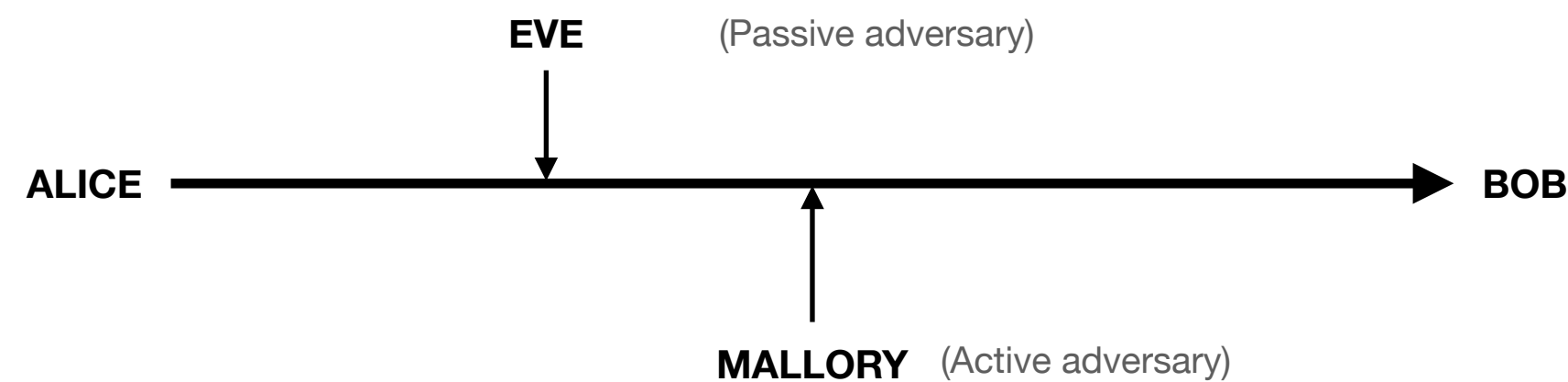
What do “domain name” and “entity” even mean?

google.com	
amazon.com	
snapchat.com	
twitter.com	

Binding of domain name to entity

Cryptography

Authentication Requirements

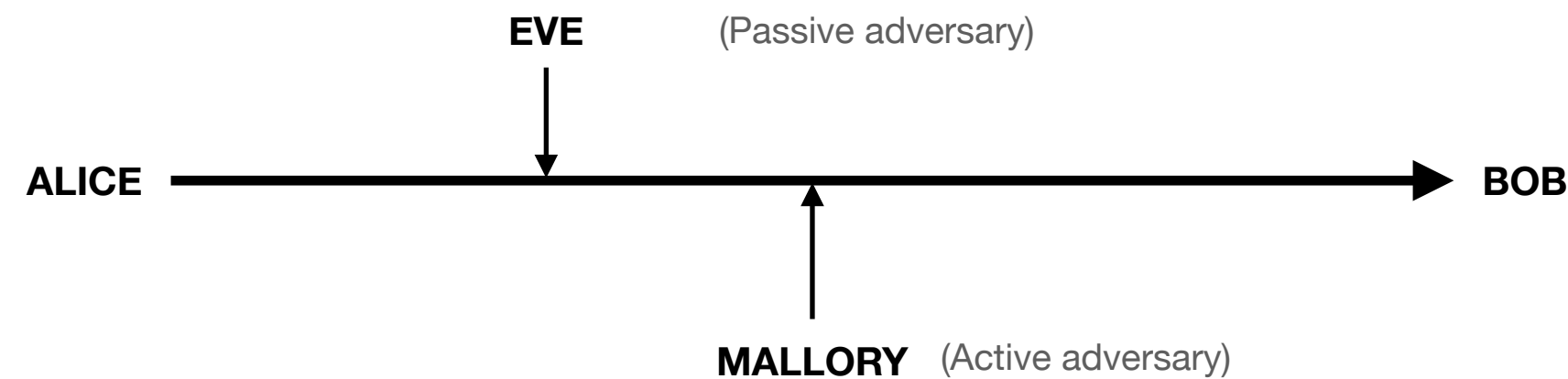


In the context of communication across a network, the following attacks can be identified:

- **Disclosure** – releases of message contents to any person or process not possessing the appropriate cryptographic key.
- **Traffic analysis** – discovery of the pattern of traffic between parties.
- **Masquerade** – insertion of messages into the network fraudulent source.
 - ◆ Content modification – changes to the content of the message, including insertion deletion, transposition and modification.
 - ◆ Sequence modification – any modification to a sequence of messages between parties, including insertion, deletion and reordering.
- **Timing modification** – delay or replay of messages.
- **Source repudiation** – denial of transmission of message by source.
- **Destination repudiation** – denial of transmission of message by destination.

Cryptography

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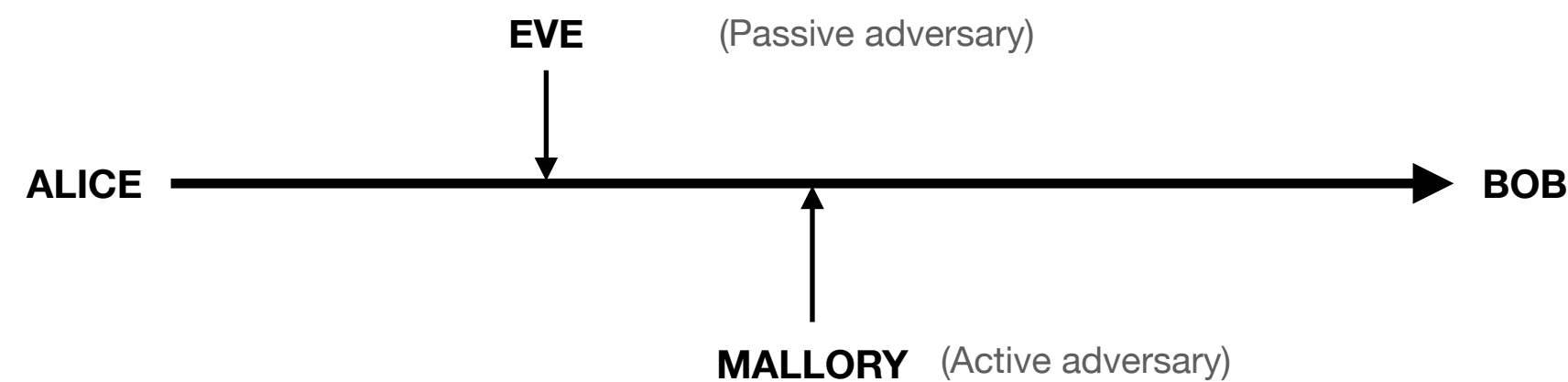


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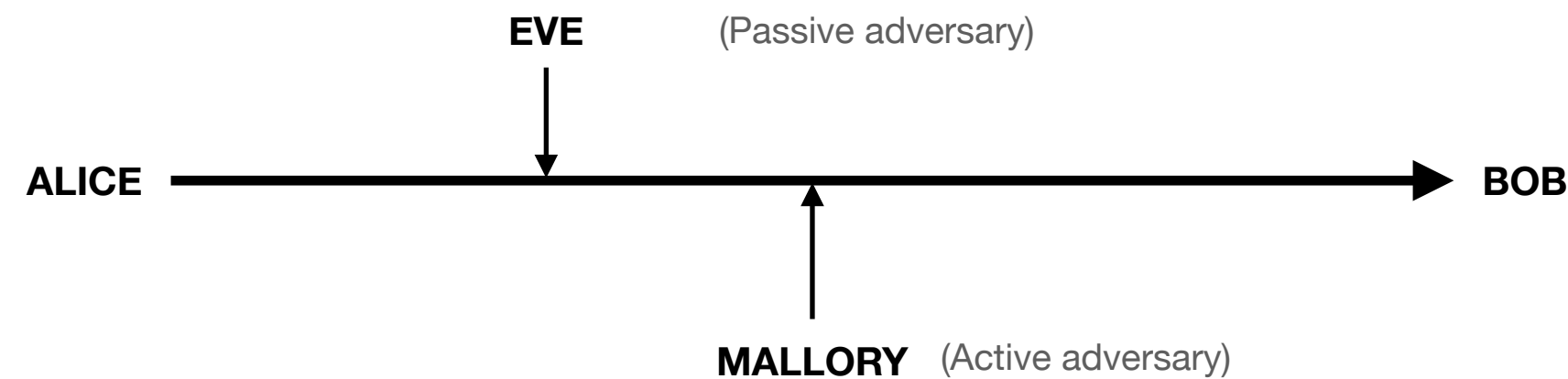
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Confidentiality

Authentication

Cryptography

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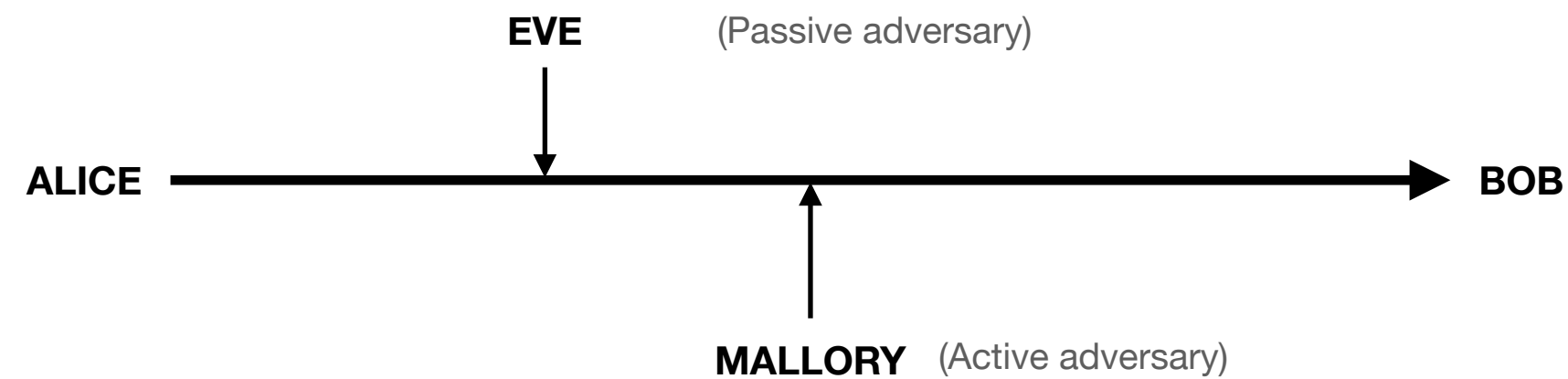


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- } **Confidentiality**
 } **Authentication**
 } **Digital signature**

Cryptography

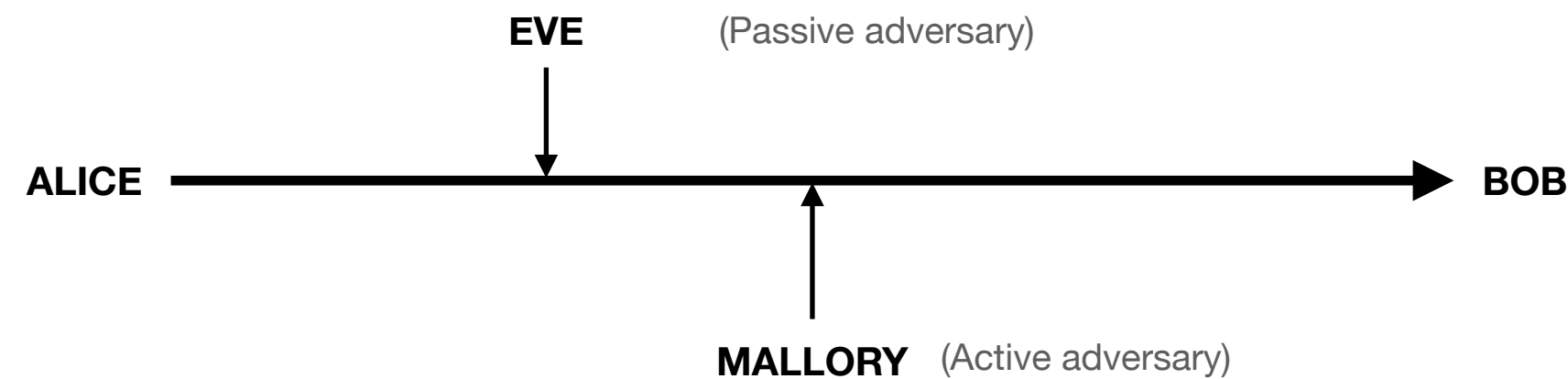
DIGITAL SIGNATURES



How can Bob verify that the message he got was from Alice. Anyone with Bob's public key can encrypt the message. Also, if Alice ever sends a message and later claims to have not sent the message, how do we verify that she is the one who sent the message? It is also possible that Mallory can alter the message from Alice, how do verify that the message has not been changed. Asymmetric cryptography can help us to provide these proofs through digital signatures.

Cryptography

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A digital signature can

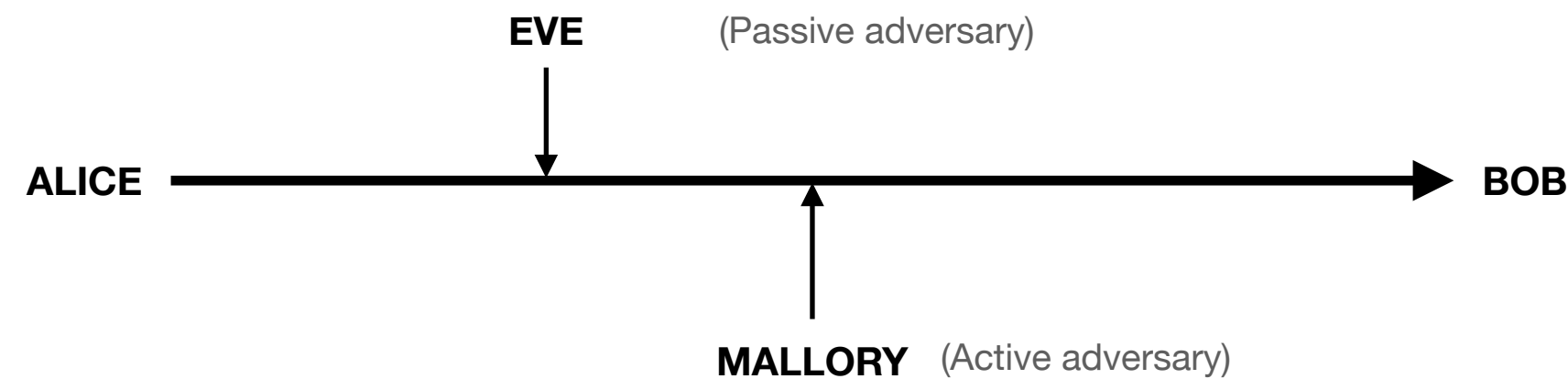
Verify the sender: *It can confirm the identify of the person/system from whom the message originated from.*

Prevent the sender from disowning the message: *it can also be used to prevent the sender from disowning the message.*

Prove the integrity of the message: *A signature can also prove that the message has not been altered/temptered after it was signed.*

Cryptography

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Prevent the sender from disowning the message: it can also be used to prevent the sender from disowning the message.

Prove the integrity of the message: A signature can also prove that the message has not been altered/temptered after it was signed.

Thus, a good digital signature should be **tied to the message and the signer, they should never be separated**. They should be also easy to be verified by other parties. Digital signature schemes therefore have to distinct steps: **the signing process and the verification process**.

Cryptography

DIGITAL SIGNATURES

	Confidentiality	Integrity/Authentication
Symmetric keys	Symmetric encryption	Message Authentication Codes (MAC)
Asymmetric	Asymmetric encryption	Digital signatures

Cryptography

DIGITAL SIGNATURES

Any message authentication or digital signature mechanism can be viewed as having fundamentally two levels.

Lower level - A function that produces an authenticator: A value to be used to authenticate a message

Higher level - Protocol that uses the lower level primitive to enable the receiver to verify the authenticity of a message.

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The different types of functions that may be used to produce an authenticator are as follows:

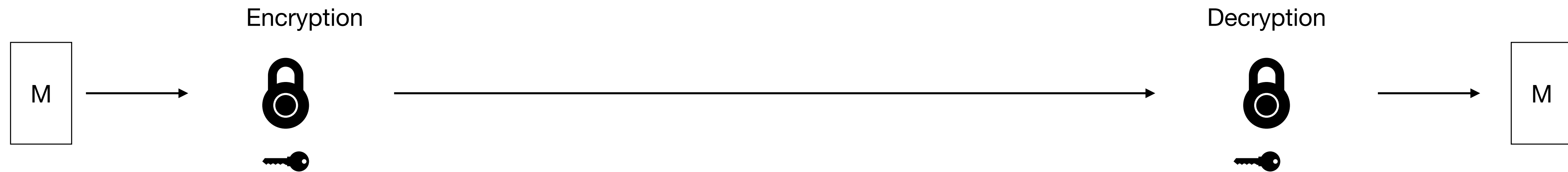
Message encryption – the cipher text of the entire message serves as its authenticator.

Message authentication code (MAC) – a public function of the message and a secret key that produces a fixed length value serves as the authenticator.

Hash function – a public function that maps a message of any length into a fixed length hash value, which serves as the authenticator.

Cryptography

DIGITAL SIGNATURES

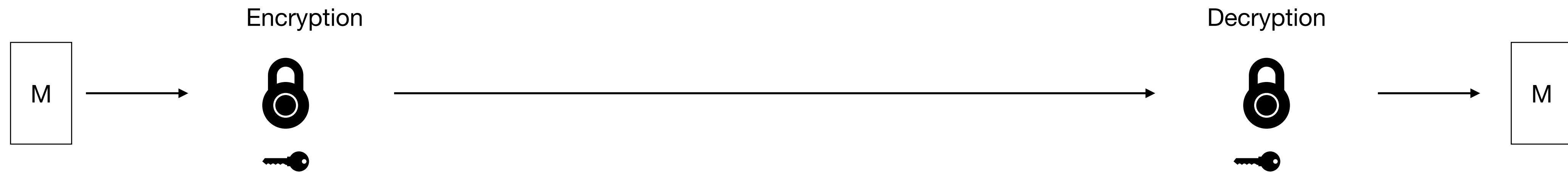


Symmetric Encryption: Confidentiality and authentication

No one can recover the plaintext of the message unless they have been given the keys.

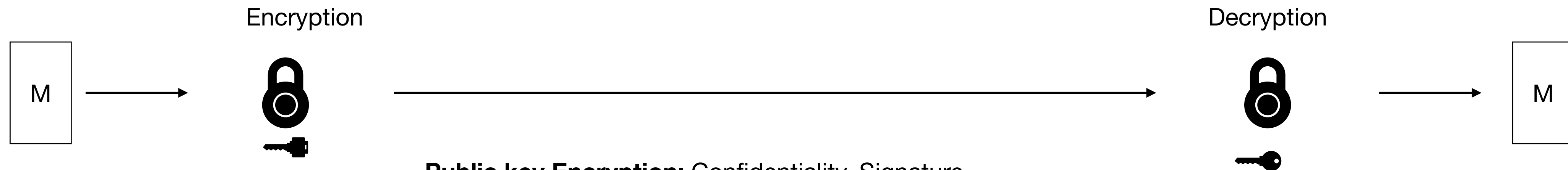
Cryptography

DIGITAL SIGNATURES



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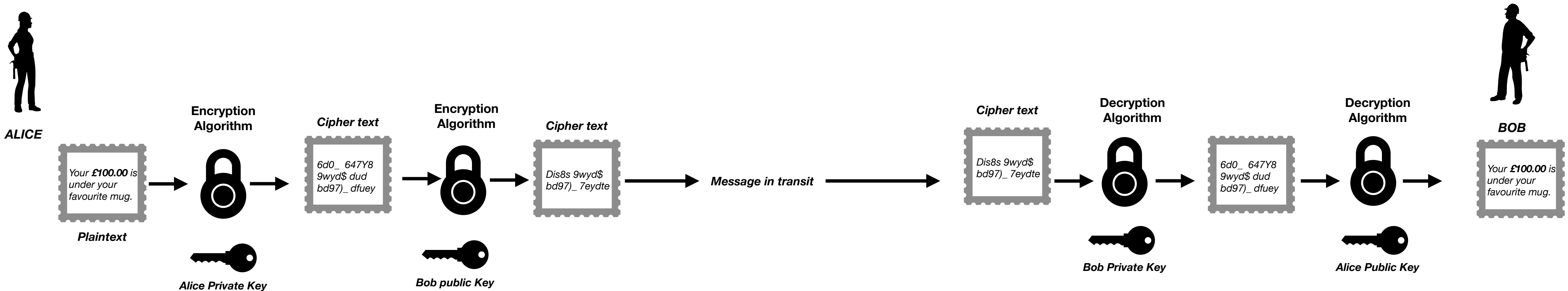


Public key Encryption: Confidentiality, Signature

The receiver with the right private key is the only one who can decrypt the message.

Cryptography

DIGITAL SIGNATURES



Public key Encryption: Confidentiality, Authentication, Signature

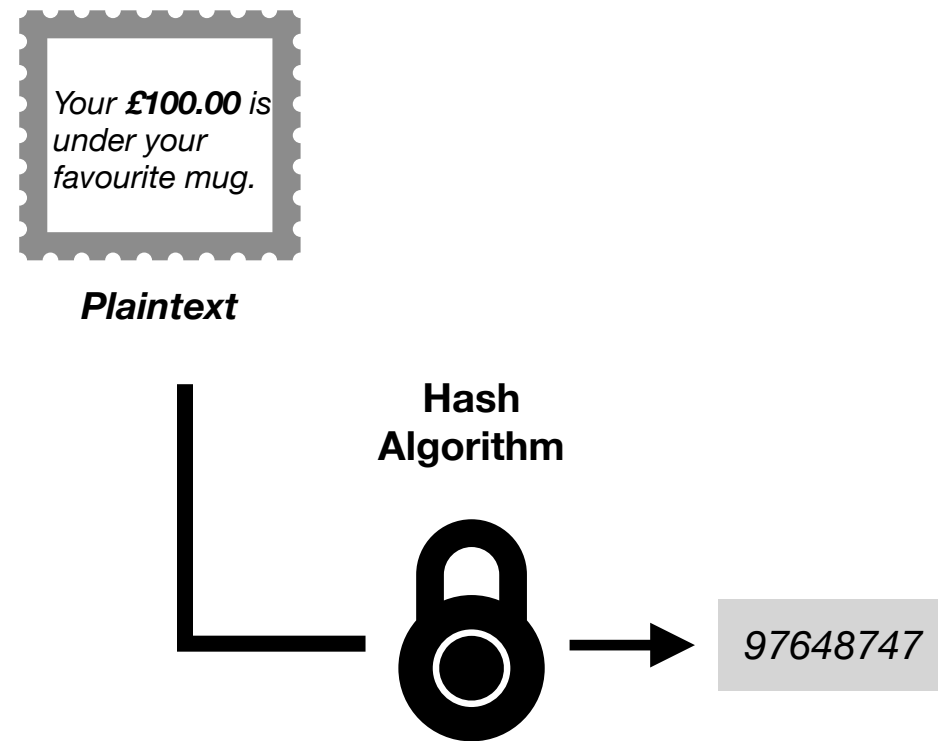
1. After creating a message, Alice encrypts the message with her own private key.
2. Then, she encrypts the generated ciphertext with Bob's public key.
3. Alice sends both the encrypted message to Bob.
4. When Bob receives them, he decrypts the ciphertext using his own private key, and then uses Alice's public key to decrypt the message. If he cannot decrypt the ciphertext with Alice's public key, he will know that the message is not from Alice.
5. Bob can be confident that the message was from Alice and was intended to come to him.

Cryptography

DIGITAL SIGNATURES



ALICE



The steps for Alice to send a digitally signed message to Bob are as follows:

1. After creating a message, Alice generates a digest on it.

Cryptography

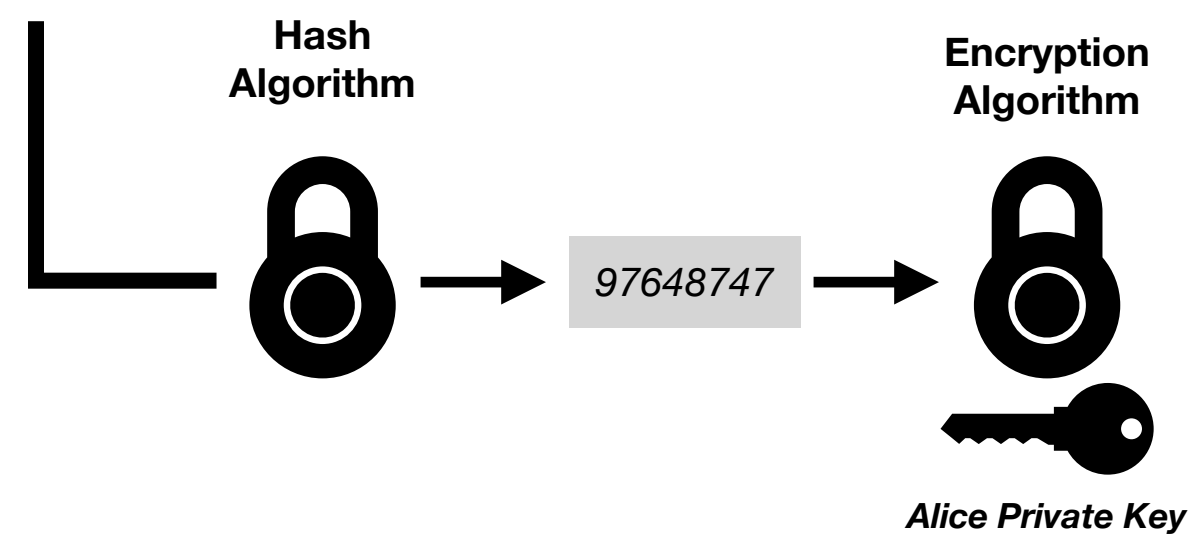
DIGITAL SIGNATURES



ALICE



Plaintext



The steps for Alice to send a digitally signed message to Bob are as follows:

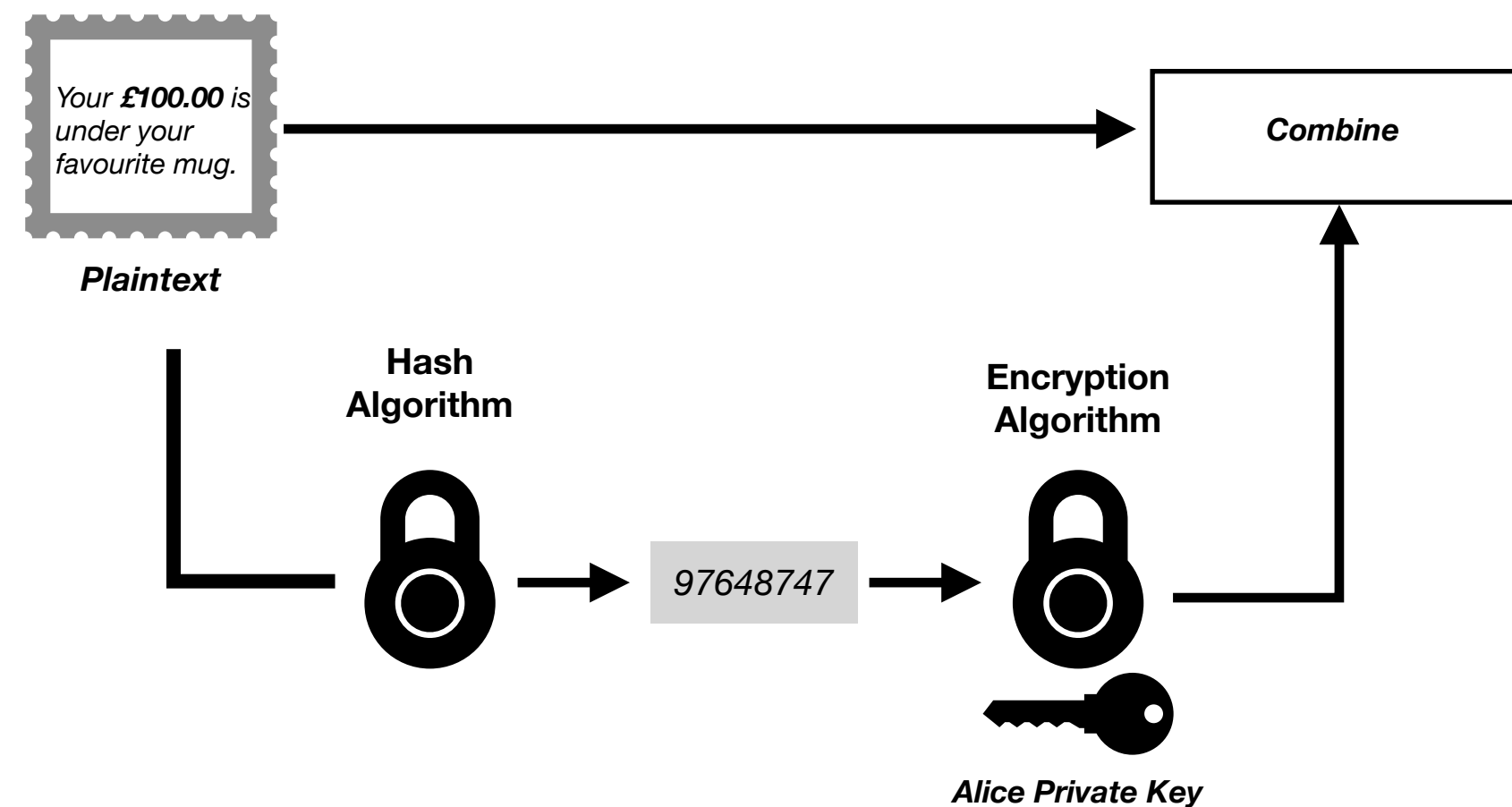
1. After creating a message, Alice generates a digest on it.
2. Alice encrypts the digest with her private key. The encrypted digest is the digital signature for the message.

Cryptography

DIGITAL SIGNATURES



ALICE



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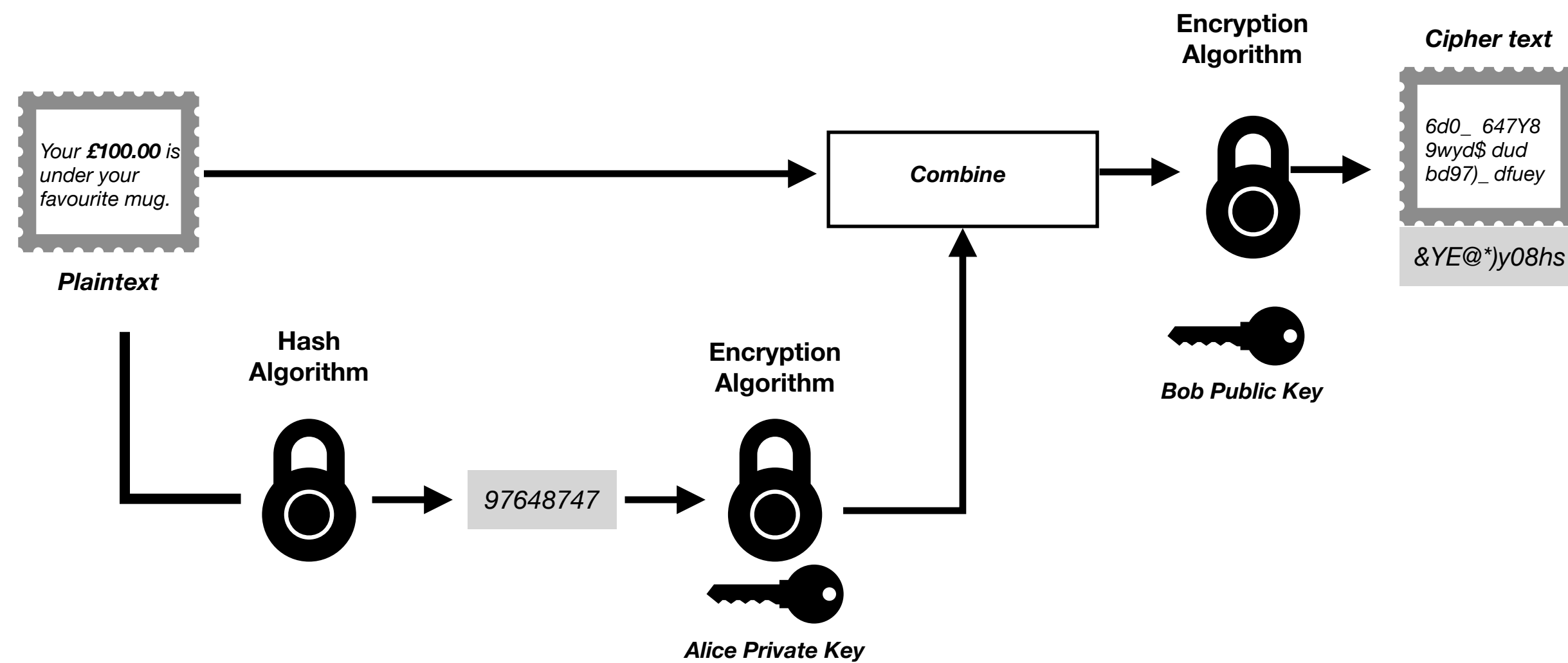
1. After creating a message, Alice generates a digest on it.
2. Alice encrypts the digest with her private key. The encrypted digest is the digital signature for the message.
3. Alice combines both the message and the digital signature.
4. When Bob receives them, he decrypts the digital signature using Alice's public key, revealing the digest. If he cannot decrypt the digital signature, then he knows that it did not come from Alice (because only Alice's public key can decrypt the digest generated with her private key).
5. Bob then hashes the memo with the same hash algorithm Alice used and compares the result to the digest he received from Alice. If they are equal, Bob can be confident that the message has not changed since she signed it. If the digests are not equal, Bob will know the message has changed since it was signed.

Cryptography

DIGITAL SIGNATURES



ALICE

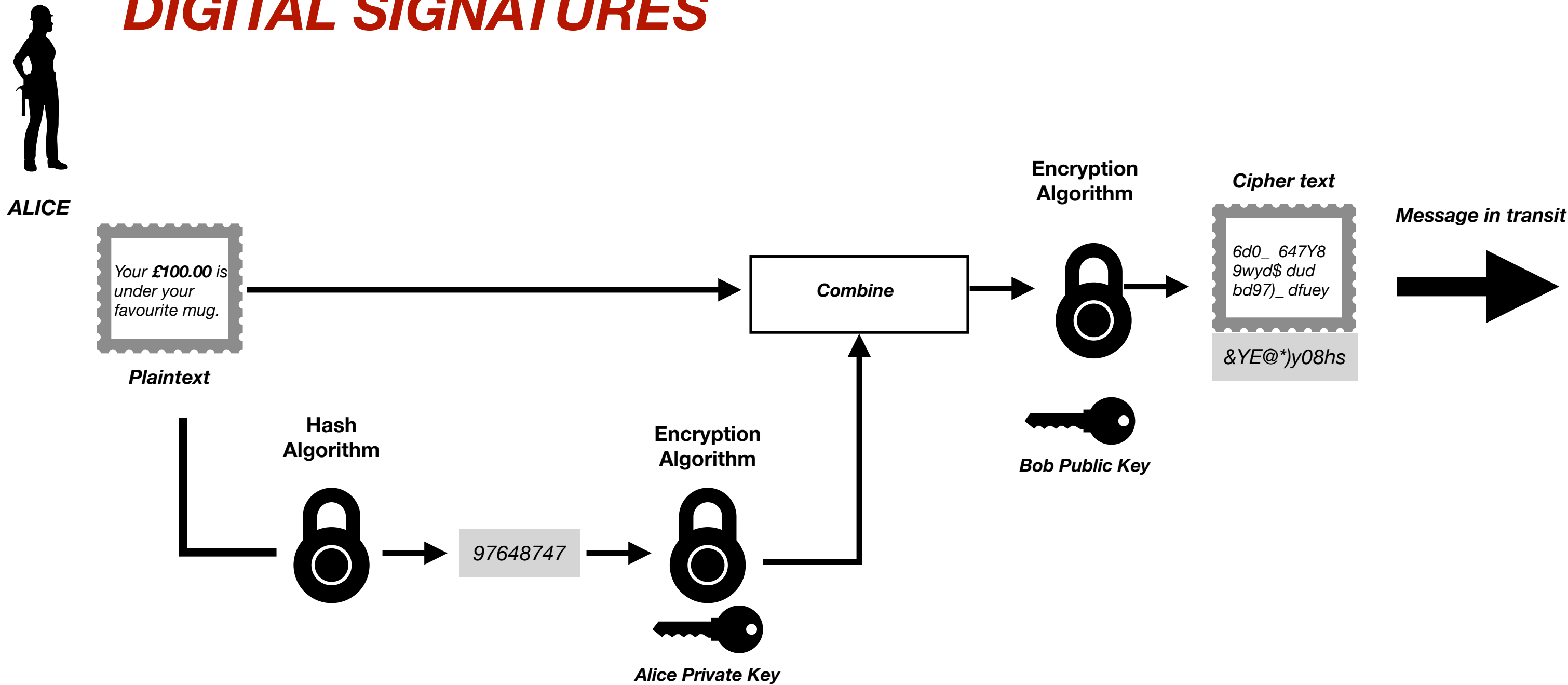


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1. After creating a message, Alice generates a digest on it.
2. Alice encrypts the digest with her private key. The encrypted digest is the digital signature for the message.
3. Alice combines both the message and the digital signature.
4. Alice will now use Bob's public key to ensure confidentiality of the message.

Cryptography

DIGITAL SIGNATURES

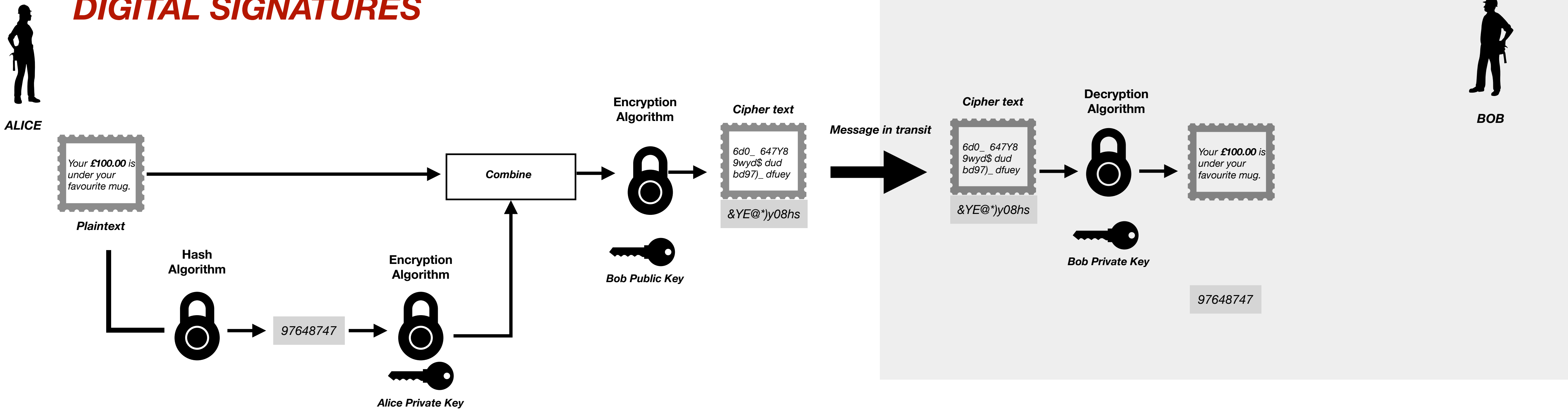


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Cryptography

DIGITAL SIGNATURES

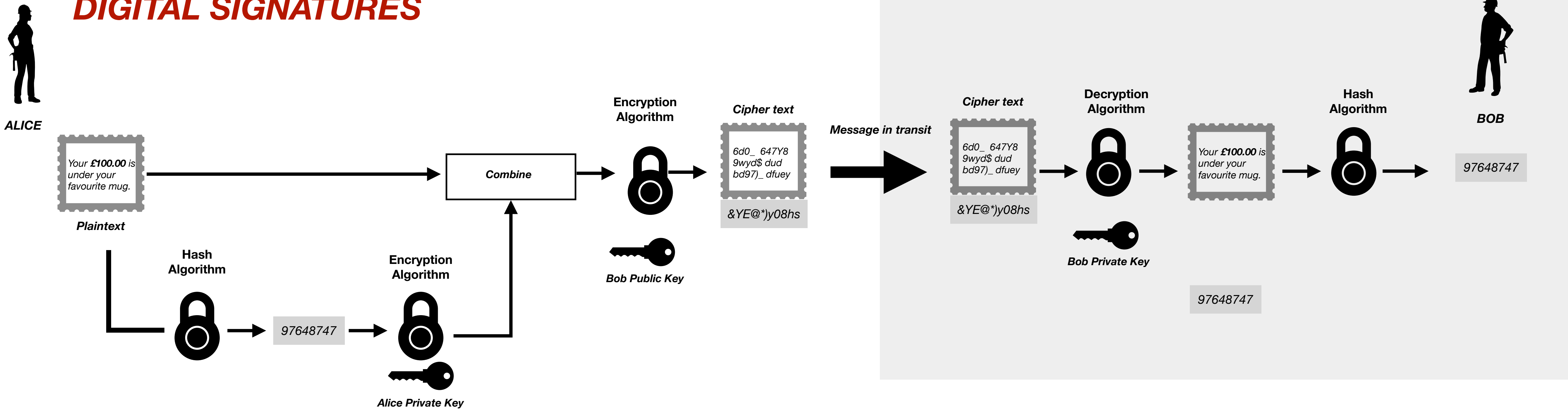


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Cryptography

DIGITAL SIGNATURES



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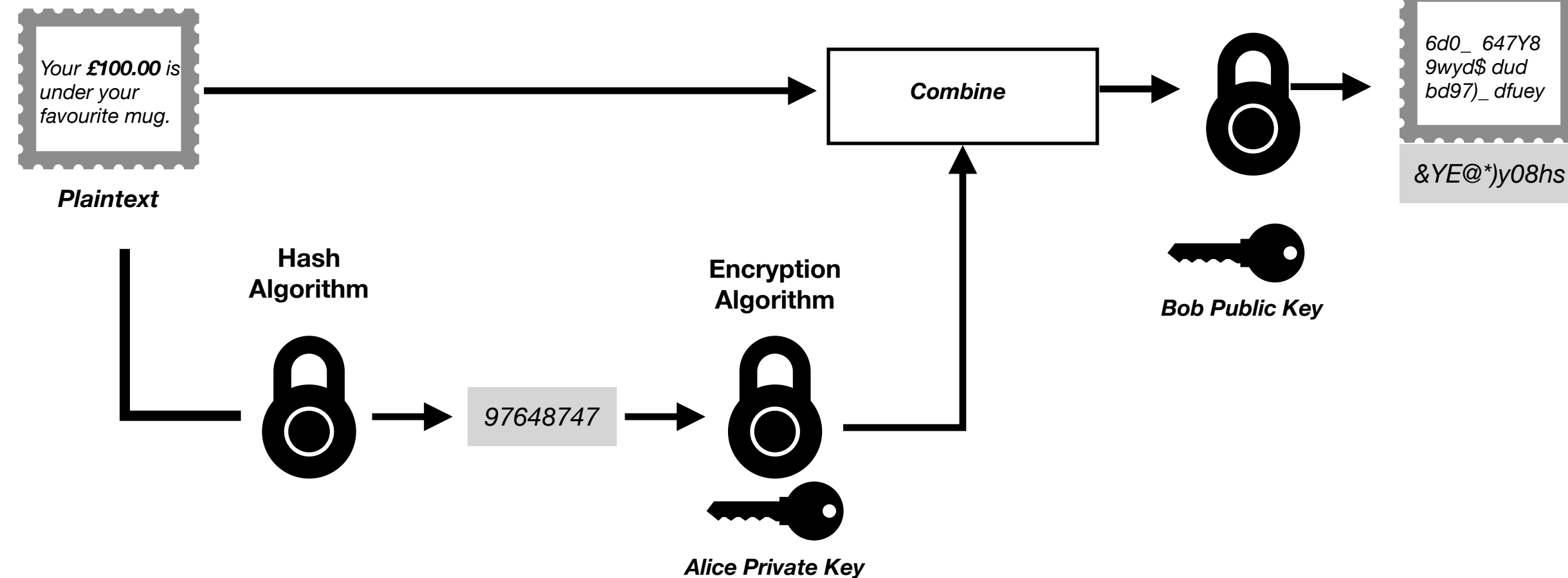
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6. Bob would then use the message to produce a hash using the same algorithm that Alice used.

Cryptography

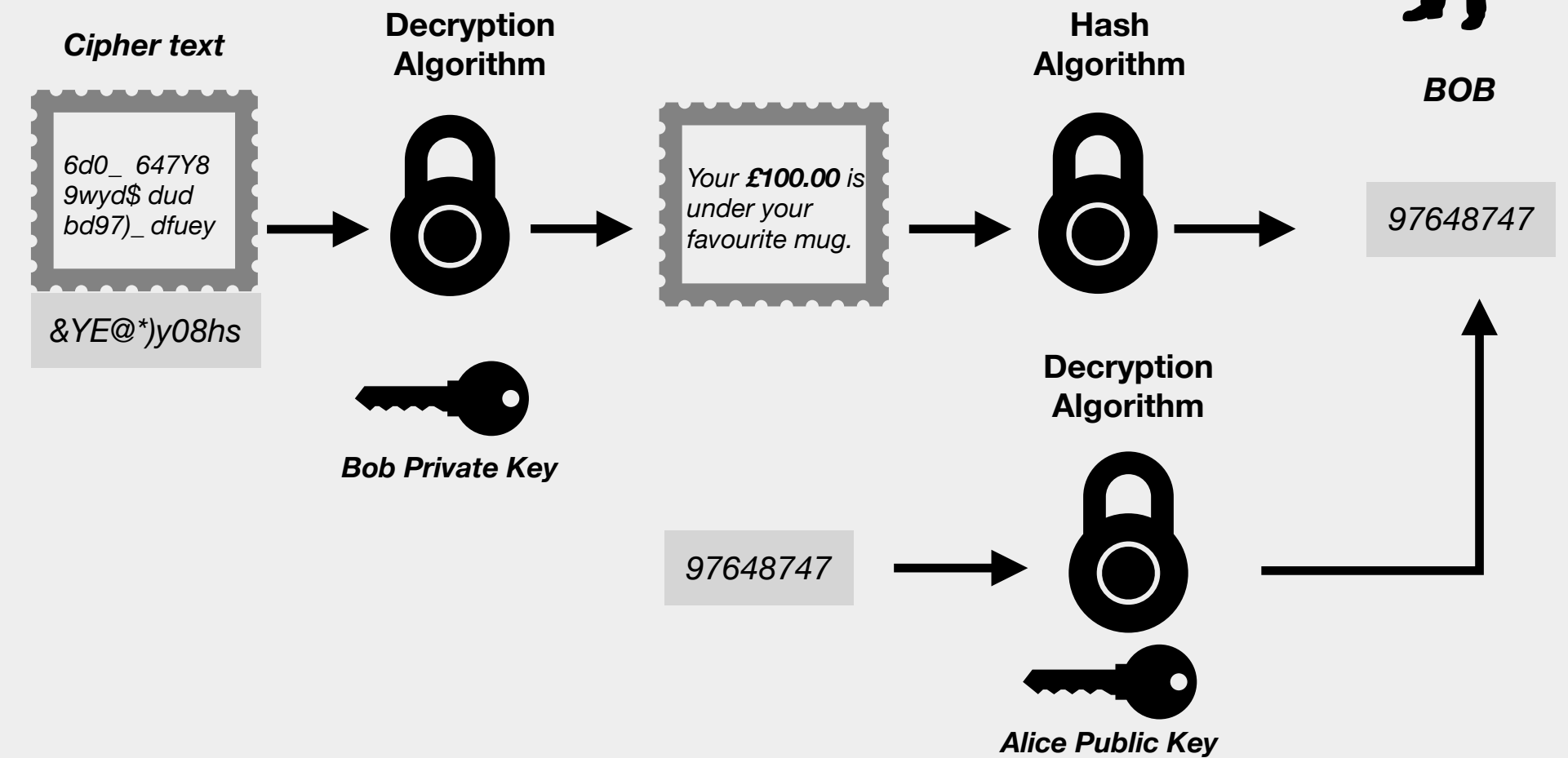
DIGITAL SIGNATURES



ALICE



Message in transit



BOB

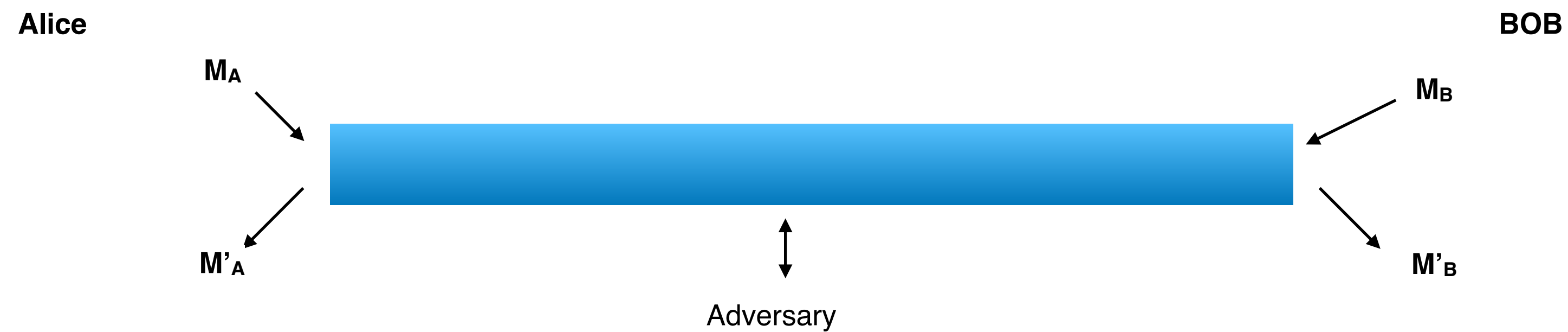
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5. When Bob receives them, he decrypts the package using his own private key to have both the message and the encrypted hash.
6. Bob would then use the message to produce a hash using the same algorithm that Alice used.
7. Bob then decrypts the encrypted hash using Alice's public key. If its successful, he knows its from Alice. But to ensure the message has not changed the compares his own hash and the hash that Alice sent over.

Cryptography

SO FAR - We have achieved confidentiality and Authenticity

The basic tasks of cryptography is establishing a secure channel. We want to do it and do it right.



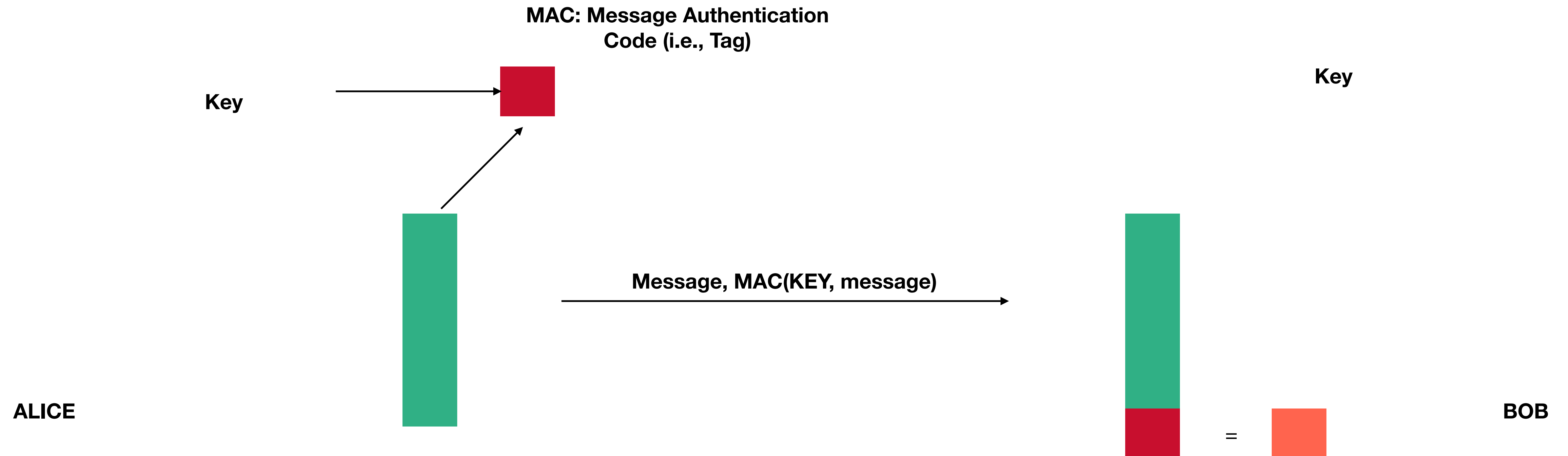
Privacy: Adversary does not learn anything about M_A and M_B

Authenticity: $M'_A = M_A$ and $M'_B = M_B$

How about Integrity?

Cryptography

SO FAR - We have achieved confidentiality and Authenticity

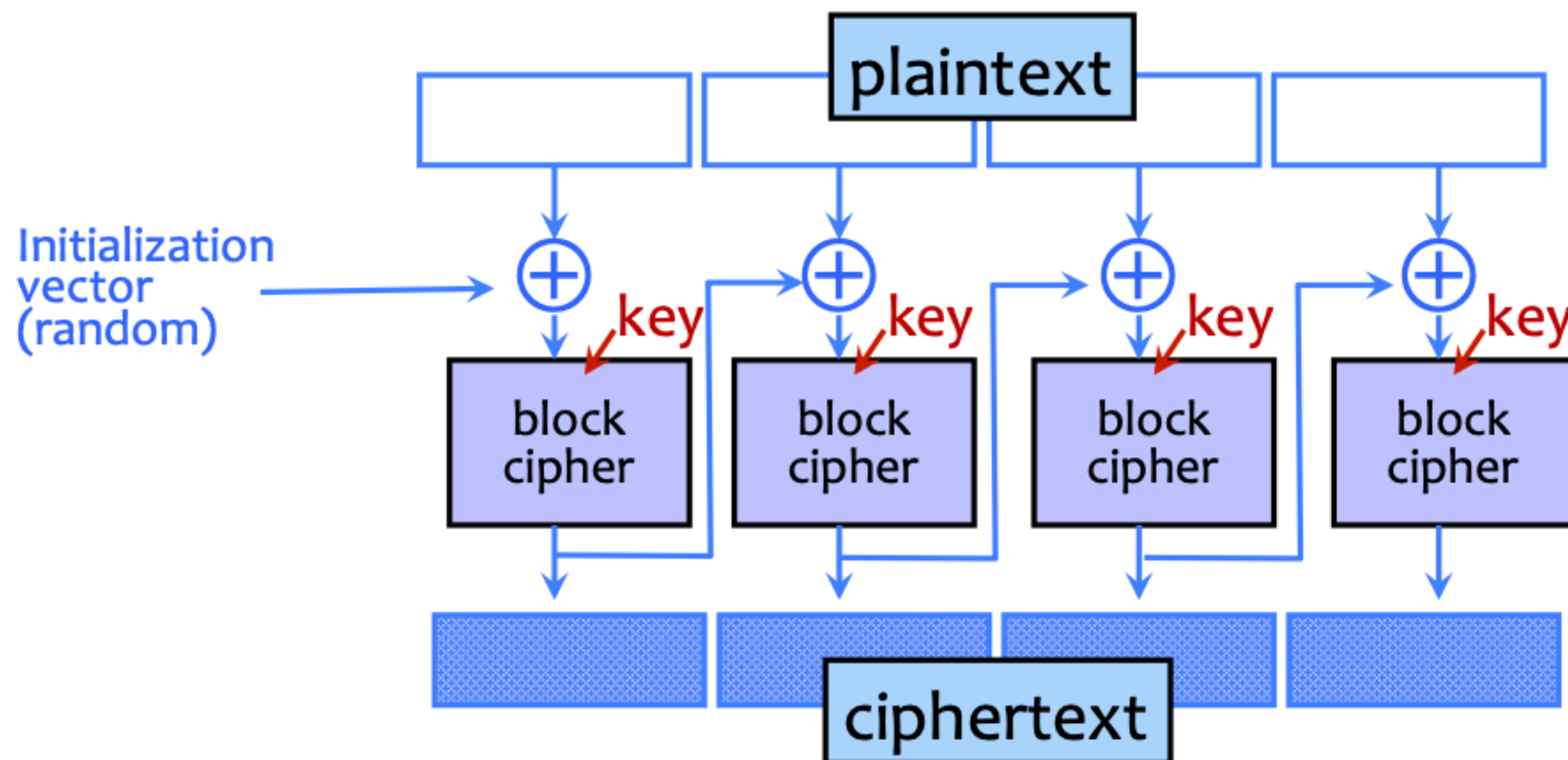


Integrity and authentication: only someone who knows KEY can compute correct MAC for a given message.

Recomputes MAC and verifies whether it is equal to the MAC attached to the message

Cryptography

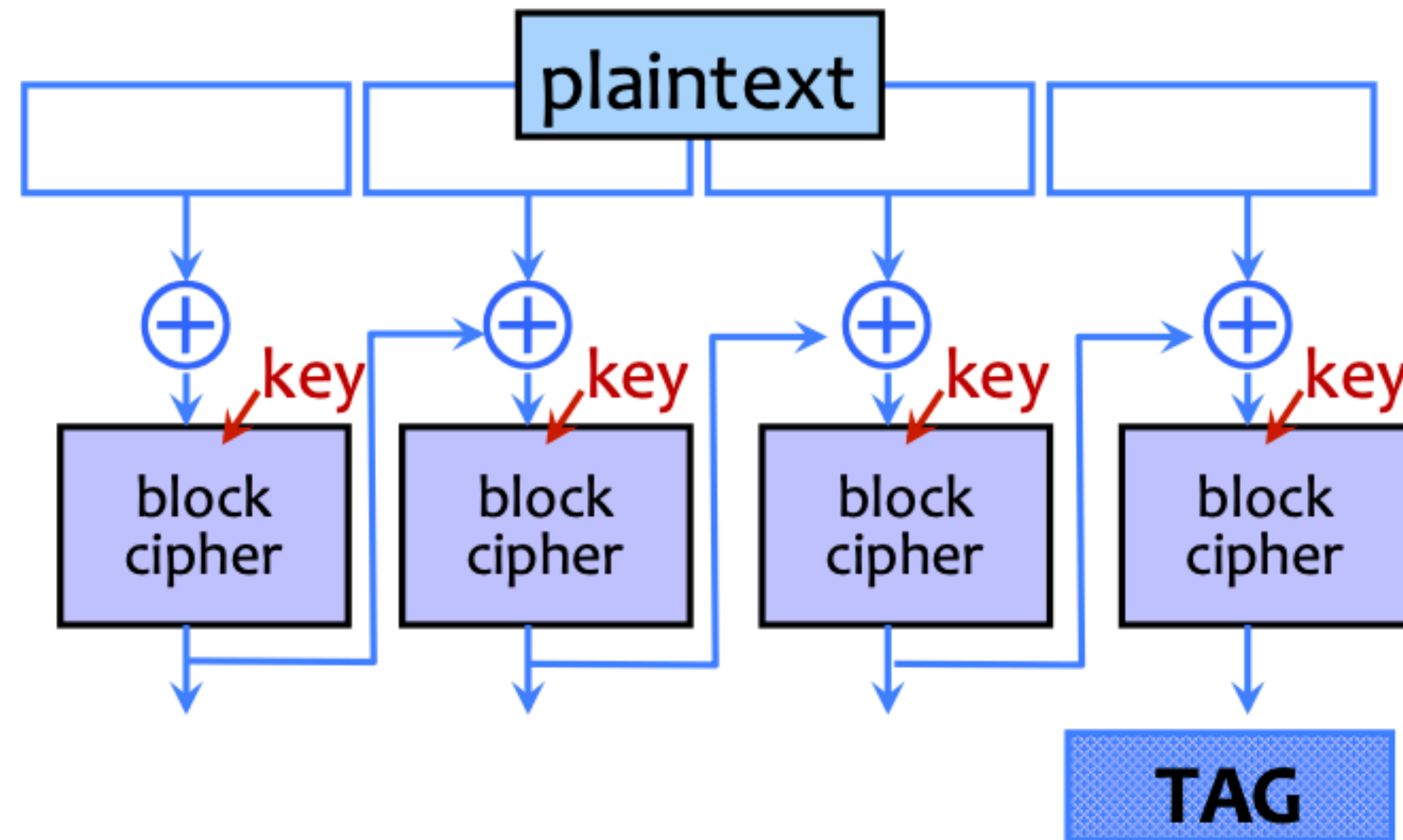
Remember: CBC Mode Encryption



- Identical blocks of plaintext encrypted differently
- Last cipherblock depends on entire plaintext
 - Still does not guarantee integrity

Cryptography

Remember: CBC - MAC



- Not secure when system may MAC messages of different lengths.
- NIST recommends a derivative called CMAC [FYI only]

Cryptography

DIGITAL SIGNATURES: MAC

An alternative authentication technique involves using a secret key to generate a small fixed size block of data called **Checksum** or **MAC**.

MAC (Message Authentication Code)

Goal:

- Authenticate the origin of the message
- Prove integrity of the message
- Non-repudiation of origin (dispute resolution)

When user A wants to send to user B, they calculate the MAC as a function of the message and the key.

Cryptography

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MAC = C_K(M), where

M = Input message

C = MAC generation function

K = shared secret key

MAC = Authenticator (Cryptographic checksum)

The MAC and Message are transmitted to the intended recipient.

To verify the message the receiver also performs the same calculation using the agreed secret key.

Cryptography

DIGITAL SIGNATURES: MAC

When user Alice wants to send to user Bob, they calculate the MAC as a function of the message and the key.

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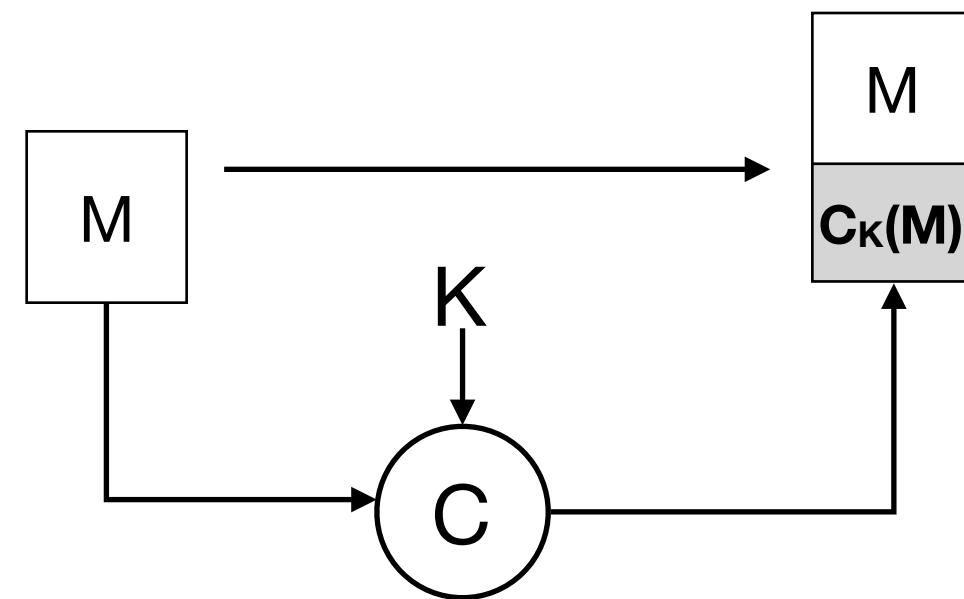
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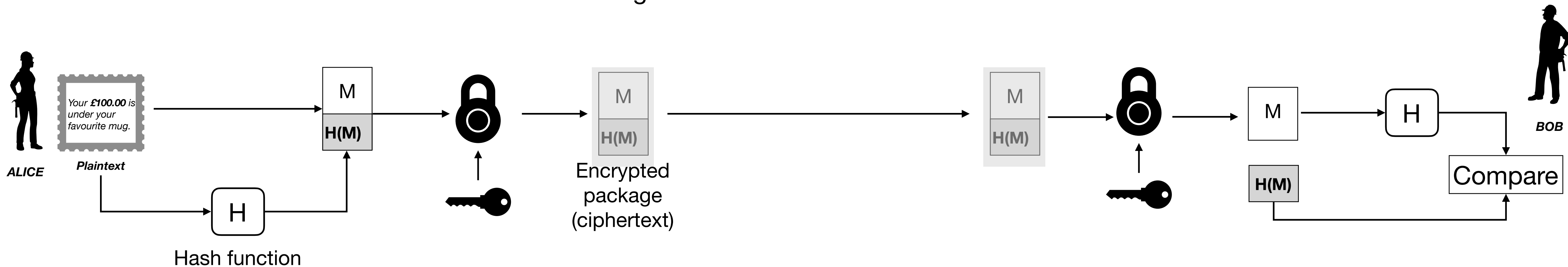
If the Alice and Bob are the only ones who know the secret key, then

- Bob is assured that the message has not been altered. If an attacker alters the message but does not alter the MAC, then the receiver's calculation of the MAC will differ from the received MAC.
- The receiver is assured that the message is from an authentic sender. Because no one else knows the secret key, no one else could prepare a message with a proper MAC

Cryptography

DIGITAL SIGNATURES: HMAC

When user Alice wants to send to user Bob using Hash MAC



There are several ways in which Alice can sign the message:

Alice can hash the message then combine the digest/hash with the message before applying encryption. In this way, the ciphertext contains the message and the hash. Bob using the right key will decrypt the message, calculate the digest/hash then compare his hash with the one that came with the message.

The second way, Alice can calculate the hash and encrypt the hash only before sending both the encrypted hash and the message to Bob. This is useful when the confidentiality of the message is not important but integrity is important. Symmetric encryption is usually used

Cryptography

DIGITAL SIGNATURES

Digital signatures function much as physical signatures.

Ideally a signature should be: unforgeable, authentic, tamperproof, non-reusable, and allow no repudiation.

Public key cryptosystems facilitate creating digital signatures.